

05 - Pymongo

Aggregation Pipelines

CSCI 112 - Contemporary Databases

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Learning Objectives

- Describe the purpose and syntax of some of the most common and useful aggregation stage operators in MongoDB.
- Explain how documents flow through the aggregation pipeline and how each stage affects the document stream.
- Apply the aggregation framework to perform data analysis and transformation tasks on MongoDB collections

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Scenario for the Day



Context Setting

- Rural Savings would like to start their digital transformation with PRIME, their core banking system for retail credit card accounts.

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Aggregation Pipelines

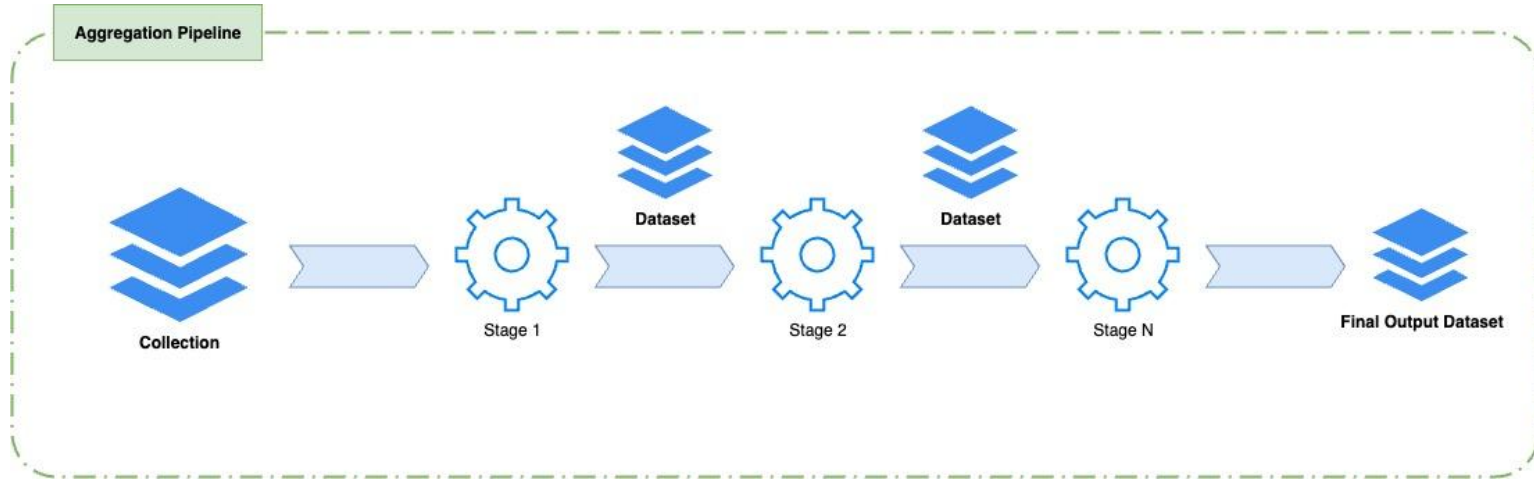


Aggregation Pipeline

- MongoDB Aggregation pipelines allow quick transformations and aggregations to collections, without exporting them into a separate analytics platform.
- Important Terms:
 - ▷ Pipeline - A set of steps or instructions to process data.
 - ▷ Stages - Each step/instruction in an aggregation pipeline is called a stage.



Aggregation Pipeline





Running Aggregation Pipelines

- Like MongoDB CRUD functions, aggregations pipelines can be run using mongosh, MongoDB Canvas, and Pymongo.
 - ▷ mongosh and Pymongo use the same function: `.aggregate()`
 - ▷ MongoDB Canvas provides a user interface to build an aggregation pipeline.



Aggregation Pipeline Function in mongosh

```
db.collection.aggregate([ <insert your pipeline here>])
```

mongosh uses the same syntax with CRUD functions, but the actual function is called `aggregate()`. This function accepts an array of JSON formatted objects, since every stage in an aggregation pipeline is a JSON formatted object.



Aggregation Pipeline Function in Pymongo

```
from utils import *

def myFirstAggPipeline():
    conn = openConnection()
    db = conn['ruralSavingsPrime']

    pipeline = [] # a pipeline is a list of dictionaries

    db['transactions'].aggregate(pipeline)
```

In pymongo, it's the same function. However, the best practice is to store the aggregation pipeline into a dictionary list for better readability.



Aggregation Pipeline Creation

- To dive deep into the creation of MongoDB Aggregation Pipelines, go to the course pack and navigate to lecture_supplements > 05-aggregation-pipelines

End.

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